

# Curriculum Vitae

Gustavo Marques de Oliveira Martins

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## Personal Information

Name: Gustavo Marques de Oliveira Martins

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## Address

IMPA - Instituto Nacional de Matemática Pura e Aplicada

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## Education

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| 2023-2027<br>(expected) | <b>Mathematics Ph.D.</b><br>IMPA - Instituto de Matemática Pura e Aplicada, Rio de Janeiro/RJ Brazil<br>Advisor: Lucas Ambrozio<br>Visiting Student at University of Chicago (2025-2026)<br>Host: André Neves<br>Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)/<br>CAPES Programa Doutorado-sanduíche no Exterior (PDSE) |
| 2021-2023               | <b>Mathematics M.S.</b><br>IMPA - Instituto de Matemática Pura e Aplicada, Rio de Janeiro/RJ Brazil<br>Advisor: Lucas Ambrozio<br>Scholarship: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)/<br>Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro<br>(FAPERJ)                                            |
| 2017-2022               | <b>Applied Mathematics B.S. - Emphasis in Business Mathematics</b><br>UFRJ - Universidade Federal do Rio de Janeiro/RJ Brazil                                                                                                                                                                                                                                |

## Research interests

Mathematics  
Differential Geometry  
Geometric Analysis

## Publications and Preprints

Gustavo Martins, *Topological and spectral rigidity of hypersurface Zoll manifolds*, preprint, arXiv:2609.20689 [math.DG].

## General talks

- |      |                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021 | XLII Jornada Giulio Massarani de Iniciação Científica, Tecnológica, Artística e Cultural (JICTAC 2020): “O Teorema de Mergulho de Kodaira”, UFRJ, Brazil. |
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| 2021 | XLII Jornada Giulio Massarani de Iniciação Científica, Tecnológica, Artística e Cultural (JICTAC 2020): “Suavização de tábuas de mortalidade via modelos dinâmicos lineares”, UFRJ, Brazil. |
| 2019 | Encontros Matemáticos: “Geometria e Economia” - UFRJ, Brazil.                                                                                                                               |

### Events and Conferences

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|------|---------------------------------------------------------------------------------------------|
| 2026 | XXII Escola Brasileira de Geometria Diferencial (participant).                              |
| 2026 | 2026 Lake Michigan Workshop on Geometric Analysis Seminar (participant)                     |
| 2025 | 35 <sup>o</sup> Colóquio Brasileiro de Matemática (participant).                            |
| 2025 | Problem-solving Workshop in Computational Geometric Analysis (participant)                  |
| 2025 | Systolic and Diastolic Geometry (participant).                                              |
| 2024 | Special Holonomy and Geometric Structures on Complex Manifolds (participant)                |
| 2023 | 34 <sup>o</sup> Colóquio Brasileiro de Matemática (participant).                            |
| 2022 | Conference IMPA 70 years (participant).                                                     |
| 2021 | 33 <sup>o</sup> Colóquio Brasileiro de Matemática (participant).                            |
| 2020 | 3 <sup>a</sup> Semana da Engenharia Matemática e Matemática Aplicada da UFRJ (participant). |
| 2019 | 32 <sup>o</sup> Colóquio Brasileiro de Matemática (participant).                            |

### Teaching Experience

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|-----------------------------------------------|---------------------------------------------------------------------------------------------|--------|
| IMPA                                          | Teaching Assistant<br>Partial Differential Equations                                        | 2023   |
| UFRJ                                          | Teaching Assistant (ran problem sessions), MAE240,<br>Real Analysis.                        | 2019.2 |
| Pontifícia Universidade<br>Católica (PUC-Rio) | Teaching Assistant (ran problem sessions), ECO1721,<br>Introduction to Economic Statistics. | 2018.2 |
|                                               | Teaching Assistant (ran problem sessions), ECO1109,<br>Introduction to Economics I.         | 2018.1 |

### Students and mentoring

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| University of Chicago | Directed Reading Program (DRP) mentor<br>Student mentored: Josh Johnson<br>Topic: Minimal surfaces and the Plateau Problem<br>Description: The goal was to study the solution to the Plateau Problem by Douglas and Rado as explained in Chapter 4 of T. H. Colding and W. P. Minicozzi II's book <i>A Course in Minimal Surfaces</i> . | 2026 |
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### Honours and Awards

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|------|-----------------------------------------------------------------|
| 2025 | Bolsa CAPES do Programa Doutorado-sanduíche no Exterior (PDSE). |
| 2022 | Bolsa Nota 10, FAPERJ.                                          |

2021 Best project presented in the Mathematics field at XLII Jornada Giulio Massarani de Iniciação Científica, Tecnológica, e Cultural (JICTAC 2020 - Edição Especial).

### Other Activities

2020-2021 Iniciação Científica, UFRJ  
Advisors: Mariane Branco Alves, Thais Cristina Oliveira da Fonseca, Viviana das Graças Ribeiro Lobo  
Description: the project was done at the Laboratório de Matemática Aplicada (LabMA) with funding from Mongeral Aegon. The goal was to apply Dynamical Linear Models in the study and smoothing of actuarial life tables.

2018-2021 Iniciação Científica, UFRJ  
Advisor: Andrew Clarke  
Description: the project was intended as an introduction to differential and complex geometry, with an emphasis in understanding the Kodaira Embedding Theorem.

2017-2018 Attended classes in the Economics Department at PUC-Rio. This includes courses in Micro and Macroeconomics, Public Sector Economics, Monetary Economics and Econometrics.